1a Given $T = tu(t)$, we need to find $T'$.

From derivative formula:

$$\int_{-\infty}^{+\infty} T'(t)\cdot x(t)\,dt = -\int_{-\infty}^{+\infty} T(t)\cdot x'(t)\,dt$$

$$= -\int_{-\infty}^{+\infty} t\cdot u(t)\cdot x'(t)\,dt$$

$$= -\int_{0}^{\infty} t\cdot x'(t)\,dt = -\int_{0}^{\infty} \underbrace{t}_{u}\cdot \underbrace{d(x(t))}_{dv}$$

$$= -\left[t\cdot x(t)\right]_{0}^{\infty} + \int_{0}^{\infty} x(t)\,dt$$

Since $x$ is Schwartz: $t\cdot x(t) \to 0$ as $t \to +\infty$

Also $t\,x(t) = 0$ at $t = 0$

$$\therefore \int_{-\infty}^{+\infty} T'(t)\,x(t)\,dt = \int_{0}^{\infty} x(t)\,dt$$

$$= \int_{-\infty}^{+\infty} u(t)\,x(t)\,dt = \langle u, x\rangle$$

Thus, we conclude that $T' = u$ as a distribution.

1b $T = |t|$

$$\langle T', x\rangle = -\int_{-\infty}^{+\infty} T(t)\cdot x'(t)\,dt$$

$$= -\int_{-\infty}^{+\infty} |t|\, x'(t)\,dt$$

$$= -\int_0^{\infty} t \cdot x'(t)\,dt + \int_{-\infty}^{0} t\, x'(t)\,dt$$

$$= -\left[t\,x(t)\right]_0^{\infty} + \int_0^{\infty} x(t)\,dt$$

$$+ \left[t\,x(t)\right]_{-\infty}^{0} - \int_{-\infty}^{0} x(t)\,dt$$

Again, use the fact that $x(t)$ is Schwartz.

$$= \int_0^{\infty} x(t)\,dt - \int_{-\infty}^{0} x(t)\,dt$$

$$= \int_{-\infty}^{+\infty} \operatorname{sgn}(t) \cdot x(t)\,dt$$

$$= \langle \operatorname{sgn}, x \rangle.$$

$$\therefore \quad T' = \operatorname{sgn}.$$

1e $\quad T = \cos t \cdot u(t).$

$$\langle T', x \rangle = -\int_{-\infty}^{+\infty} T(t) \cdot x'(t)\,dt$$

$$= -\int_0^{\infty} \underbrace{\cos t}_{u} \cdot \underbrace{d(x(t))}_{dv}$$

$$= -\left[\cos t \cdot x(t)\right]_0^{\infty} + \int_0^{\infty} x(t) \cdot (-\sin t)\,dt$$

Use the fact that $x$ is Schwartz: $x(\infty) = 0$

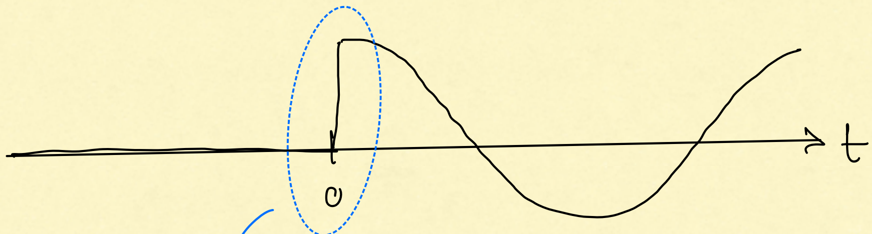
$$= -[0 - x(0)] - \int_{-\infty}^{+\infty} x(t)\cdot \sin t \cdot u(t)\,dt$$

$$= \int_{-\infty}^{+\infty} \delta(t)\, x(t)\,dt - \int_{-\infty}^{+\infty} \sin(t)\, u(t)\cdot x(t)\,dt$$

$$= \int_{-\infty}^{+\infty} [\delta(t) - \sin(t)u(t)]\, x(t)\,dt .$$

$$\Longrightarrow \quad T' = \delta(t) - \sin(t)\, u(t)$$

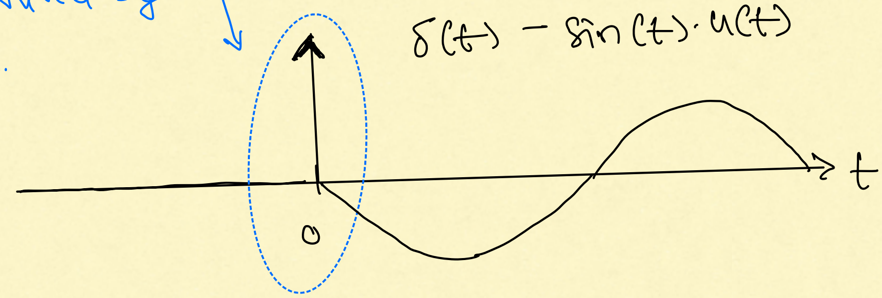
$\cos t \cdot u(t)$




The discontinuity of height 1 at time $t = 0$ is captured by $\delta(t)$.

$\Downarrow$ Differentiate

$\delta(t) - \sin(t)\cdot u(t)$




2c $\quad h(t) = \delta'(t) - 2\delta''(t) + \delta(t-2)$

2d $\quad h(t) = 1$ (= 1 for all time t)

3b

$$\int_{-\infty}^{+\infty} \delta'(t)\,dt = \langle \delta', 1 \rangle$$

$$= -\langle \delta, x' \rangle \quad \text{where } x(t) = 1 \text{ for all } t$$

$$= -x'(0)$$

$$= 0.$$

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$$\to \boxed{\delta'} \to \boxed{h} \to$$

Overall system has impulse response: $\delta' * h$

$= h'(t)$.

Note:

$$x \to \boxed{\delta'} \to \boxed{h} \to y \qquad x \to \boxed{h'} \to y$$

$$y = h * (\delta' * x)$$

$$= h * x'$$

Also: $y = h' * x$

$$\therefore \quad h * x' = h' * x$$

Also:

$$y = h * (\delta' * x) = (h * x) * \delta'$$

$$= (h * x)'$$

Thus, we have:

$$\boxed{(h * x)' = h' * x = h * x'}$$

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$$x \to \boxed{\phantom{h}} \xrightarrow{y} \boxed{h_2} \to y * h_2$$

$$h_2(t) = \sum_{k=0}^{\infty} g_k \cdot \delta(t-k)$$

$$= g_0\delta(t) + g_1\delta(t-1) + g_2\delta(t-2) + \cdots$$

$$y * h_2 = \left(x(t) + \tfrac{1}{2}x(t-1)\right) * \left(g_0\delta(t) + g_1\delta(t-1) + \cdots\right)$$

$$= x(t) * \left(g_0\delta(t) + g_1\delta(t-1) + \cdots\right)$$

$$+ \tfrac{1}{2}x(t-1) * \left(g_0\delta(t) + g_1\delta(t-1) + \cdots\right)$$

$$= g_0 x(t) + g_1 x(t-1) + g_2 x(t-2) + \cdots$$

$$+ \qquad \frac{g_0}{2}x(t-1) + \frac{g_1}{2}x(t-2) + \cdots$$

We need: $g_0 = 1$

$$g_1 + \frac{g_0}{2} = 0 \Rightarrow g_1 = -\frac{1}{2}$$

$$g_2 + \frac{g_1}{2} = 0 \Rightarrow g_2 = \left(-\frac{1}{2}\right)^2$$

$$\vdots$$

$$g_k + \frac{1}{2}g_{k-1} = 0 \Rightarrow g_k = \left(-\frac{1}{2}\right)^k$$

$$\therefore\ h_2(t) = \sum_{k=0}^{\infty}\left(-\frac{1}{2}\right)^k \delta(t-k).$$